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AUCC-224; No. of Pages 8
ARTICLE IN PRESS
Australian Critical Care xxx (2014) xxx–xxx
Contents lists available at ScienceDirect
Australian Critical Care
journal homepage: www.elsevier.com/locate/aucc

Optimal occupancy in the ICU: A literature review

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ARTICLE INFORMATION

Article history:
Received 31 July 2012
Received in revised form
12 September 2013
Accepted 26 November 2013
Available online xxx

Keywords:
Bed occupancy
Intensive care units
Critical care
Quality indicators
Health care

ABSTRACT

Introduction: In intensive care, occupancy is a commonly used measure. There is inconsistency however in its measurement and optimal occupancy targets need to be defined. The objectives of this literature review were to explore how occupancy is measured, reported, and interpreted and investigate optimal occupancy levels for ICUs.
Method: A literature search was performed using the Medline, Embase and CINAHL databases and citation tracking identified additional relevant articles. Articles published since 1997, written in English and focused on the adult ICU setting were included. As a result, 16 articles were selected for this review.
Results: Although it was apparent there was no commonly accepted or used method for calculating ICU occupancy, methods described as more accurate enumerate actual patient hours in the ICU, use operational (and preferably fully staffed) beds as the denominator, and are calculated daily. Issues pertaining to the utility, interpretation, and reporting of ICU occupancy measures were identified and there were indications that optimal ICU occupancy rates were around 70–75%. It was evident however that setting a uniform target figure for all ICUs would be problematic as there are a range of factors both at the unit and the hospital level that impact occupancy figures and optimal occupancy levels.
Implications: This literature review informed the recommendation of a proposed method for calculating ICU occupancy which provides a realistic measure of occupied bed hours as a percentage of available beds. Despite the importance of gaining an understanding of ICU occupancy at the local and broader health system levels, there are a number of unknown factors that require further research. Appropriate occupancy targets, impact of unavailable beds, and the intrinsic and extrinsic factors on occupancy measurement are a few examples of where more information is required to adequately inform ICU monitoring, planning and evaluation activities.
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Introduction

Most critically ill patients require intensive care urgently and cannot be cared for in other parts of the hospital.[1] For this reason, it is particularly important for intensive care units (ICUs) to have adequate capacity. However, clinicians and the media have depicted problems with bed shortages in Australian ICUs.[2–5] Adequate critical care service planning and evaluation, and balancing cost and efficiency with the ability to provide adequate and timely patient care require an understanding of the optimal target occupancy for an ICU.

In intensive care, occupancy is regarded as a measure of resource use, unit activity, workload, and increasingly as a quality indicator in combination with other related measures.[1,6–9] Accurate and consistent measurement of occupancy is required for the purposes of unit, hospital and health system planning, comparisons within and between health systems and providing realistic benchmarks for quality monitoring.[1,10,11] The way in which occupancy should be operationally defined, measured, interpreted and used in Australian ICUs, however, is unclear.

The aim of this literature review was to synthesise the existing literature pertaining to ICU occupancy in order to inform future quality monitoring, reporting and planning activities in ICUs in New South Wales (NSW), Australia. The specific objectives were to: (1) outline how ICU occupancy has been operationalised in the literature; (2) identify the way ICU occupancy has been and should be measured; (3) outline how measures of ICU occupancy should be reported, interpreted, and used; and (4) identify optimal occupancy levels for ICUs.

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http://dx.doi.org/10.1016/j.aucc.2013.11.003

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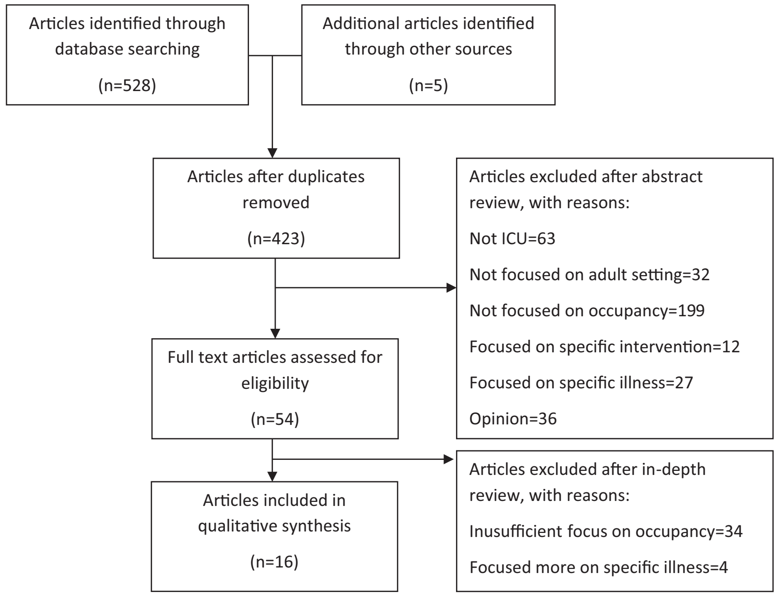

Fig. 1. Flow chart of literature review process.

Methods and materials

A literature search was performed using the Medline, Embase and CINAHL electronic databases. A combination of search terms were used including: intensive care (unit); critical care; occupancy (bed, funded and physical); and beds (available and funded). The search results were then limited to articles published between 1997 and January 2012 and written in English. A staged review process, involving an abstract review, followed by an in-depth review of each article, was used to identify relevant articles to be included in the literature review. Studies reporting occupancy in an ICU context and those with a specific focus on occupancy were included. Studies that discussed beds or bed numbers without a specific focus on occupancy or concentrated on occupancy in relation to specific ICU interventions or illnesses were excluded. Opinion articles such as editorials, letters and comments were also excluded. The numbers of identified, excluded and included articles are shown in Fig. 1.

A narrative review of the literature was then undertaken with data abstraction performed using a summary table containing key elements related to the study aims. A second rater then verified the accuracy of abstraction using the first rater's completed summary table and referring back to the original articles.

Results

A total of 16 articles included in this review are summarised in Table 1. Of these articles there were 13 research studies and three discussion papers. Of the selected articles, six were from the United Kingdom; six from North America and four from Europe.

Operationalisation

The reviewed literature lacks an operational definition of occupancy. This inadequacy has seemingly resulted in uncertainty pertaining to the purpose of measuring occupancy and what information the measure is providing. The US National Library of Medicine broadly defines occupancy as a "measure of inpatient health facility use based upon the average number or proportion of beds occupied for a given period of time".[12] The vague nature of this definition means further operationalisation is required to develop a more meaningful and practical occupancy definition that allows for consistent interpretation and comparable reporting.

Measurement

Of the 16 articles reviewed, two specifically investigated and compared commonly used occupancy calculation methods with the intention of identifying the most accurate method for ICUs.[13,14] In each of these studies, the comparison was made by applying commonly used calculation methods to the same datasets (outlined in Fig. 2). Although the reference methods used by both measured hourly ICU occupancy, they differed on the level of granularity in definition and measurement.

It was evident that the different calculation methods used produced marked differences in annual occupancy figures across individual units. This can be seen, for example, in Ridley and Rowan's comparison of methods six and seven. Method six included the entire length of stay of all patients *admitted* during the study period but no part of the stay of patients admitted just prior to the study period, while method seven (the reference method) included the stay contributed by any patients *treated* in the ICU during the study period (i.e. equivalent to occupied bed hours). Differences in the patient inclusion criteria used in these two occupancy calculation methods resulted in differences in the numbers of patient days counted and the resulting occupancy figures for the same units and time period (Figs. 3 and 4; constructed using data reported in table form by Ridley and Rowan, 1997).[13] Using only those patients admitted during the study period, method six mostly overestimated occupancy figures when compared to the reference method. The unit with the greatest difference in occupancy figures had a 77% occupancy rate according to method six and a 65% occupancy rate using method seven.[13] As can be seen in Figs. 3 and 4 however, the differences between the two methods were inconsistent across individual units. This variation was explained by the differences in patient hours included or excluded from the calculation method and the highly variable patient throughput at each ICU.

The accuracy of some occupancy calculation methods can be questioned when bed unavailability is not factored into calculations. A number of the studies included in this review[11,15,16] used data from the Hospital Cost Report Information System (HCRIS), an administrative database in the United States containing provider

Table 1
Summary of studies of ICU occupancy.

Study	Purpose	Design	Occupancy calculation	Findings	Limitations
Barado 2012 (Spain)[20]	Develop a model for simulating daily ICU OCC	9 year retrospective analysis of pt admission data and bed OCC records	–	ICU bed OCC rates depend on multiple factors e.g. scheduled surgery and pt types Mean bed OCC of 85% is misleading	Model cannot be applied to other hosp No OCC definition or calculation method
Olafson 2011 (Canada)[23]	Compare methods of describing ICU bed requirements	7 year retrospective study	Bed days Peak daily bed OCC (using admission and discharge dates and times)	Annual OCC is underestimated using the ICU bed day method compared to the peak daily OCC method	Abstract only Method of bed planning does not account for future changes in demand
Valentin 2011 (Europe)[22]	Provide recommendations for ICU planning	Literature review and consensus finding	Hourly OCC	OCC should not exceed 75%	Based solely on expert opinion
Halpern 2010 (USA)[15]	Analyse the evolving role, patterns of use and costs of CCM	5 year retrospective analysis of HCRIS data	The no. of bed days used divided by the no. of licensed bed days	CCM bed no. increased by 6.5% CCM OCC increased by 4.5%	Used "potential" bed days Inaccuracy of HCRIS data
Chrusch 2009 (Canada)[9]	Determine whether a lack of ICU beds lead to adverse outcomes i.e. premature pt discharge, subsequent early death or ICU readmission	8 year prospective cohort study	Average midnight bed census divided by the no. of funded ICU beds	High OCC rates are associated with high rates of adverse outcomes	Use of midnight bed census Other possible confounders not taken into account Biased method of determining adverse outcomes
Halpern 2006 (USA)[16]	Determine the impact of hosp size on CCM beds and OCC	16 year retrospective analysis of HCRIS data	The no. of bed days used divided by the no. of licensed bed days	Larger hosp had higher CCM OCC rates than smaller hosps No hosp in the study reached the US guidelines of 80–85% overall OCC-unrealistically high	Used "potential" bed days Inaccuracy of HCRIS data
Halpern 2004 (USA)[11]	Analyse the role, patterns of use and costs of CCM	16 year retrospective analysis of HCRIS and other data	The no. of bed days used divided by the no. of licensed bed days	CCM OCC remained constant despite increases in bed no.	Used "potential" bed days Inaccuracy of HCRIS data
Iapichino 2004 (Europe)[6]	Explore the relationship between outcome and volume of activity	4 month prospective, multi-centre, observational study	No. of pt days divided by the possible no. of nursing days in the ICU	High risk volume a more useful measure than total ICU volume High OCC associated with increased ICU and hospital mortality	Short study period Accuracy of OCC calculation method Record pt severity of illness at admission
Costa 2003 (UK)[10]	Validate a model for critical care and demonstrate its role in capacity planning	Predictions of bed capacity requirements based on a model using different bed no.'s	–	As unit size increases average OCC decreases The no. of ICU beds required will be underestimated using expected admissions, average LOS and target OCC	Use of models cannot predict future change in patterns of practice No OCC definition
Green 2002 (USA)[1]	Use queuing analysis to estimate bed unavailability in ICUs and obstetric units	Use of a queuing model to estimate delays in ICUs in New York state hosps	The ratio of occupied beds to the total no. of beds	Average OCC of 75% Average OCC for units with <5 beds was 47%	Use of out of service beds in the OCC calculation Unable to accurately model potential consequences of bed unavailability due to lack of data
Iapichino 2000 (Italy)[7]	Develop a tool for evaluating appropriateness of ICU utilisation	61 day audit of general ICUs	Pt bed days divided by available bed days	Overall OCC of 83.8% High-level OCC of 69.4% Low-level OCC of 101.1%	Long stay pts were included a no. of times
Fox 1999 (UK)[25]	Examine the impacts of opening a HDU on critical care resource use	8 week prospective study	Calculated from the exact no. of hours in the ICU recorded for each pt	Opening a HDU resulted in more appropriate use of the ICU	Effect of opening the HDU on pt flow including access and exit block unknown
Dhond 1998 (UK)[21]	Examine the workload and capacity constraints of an ICU before and after the opening of a HDU	6 month prospective study	Uses dates and times of pt admission and discharge, calculated daily	ICU OCC decreased after opening the HDU Opening the HDU generated new demand for critical care services	Confounding factors make matching of supply and demand difficult
Parker 1998 (UK)[8]	Establish whether demand for ICU services exceeded supply irrespective of increases in capacity	4 year and 9 month prospective study of pt data	The sum of the proportions of the durations of admission in each month relative to the no. of beds available	Monthly OCC is highly variable Increasing IC capacity does not result in an equivalent fall in OCC	OCC calculation does not account for bed unavailability Generalisability of results limited
Thompson 1998 (UK)[14]	Evaluate the amount of ICU resources that may be freed if a HDU were available	8 week prospective study	See Fig. 2	OCC definition used can result in large variations Hourly OCC is the "gold standard"	Location of outlying pts not detailed
Ridley 1997 (UK)[13]	Determine if different methods for calculating OCC yielded different OCC figures	Calculation of OCC from the same dataset using 7 different methods	See Fig. 2	100% OCC is unobtainable Method 7 (Fig. 2) is the most accurate method for calculating ICU OCC OCC must be interpreted with information on LOS and pt throughput	Accuracy of determining bed availability questionable Appropriateness of using mean and SD

Abbreviations: CCM: critical care medicine; HCRIS: Hospital Cost Report Information System; HDU: high dependency unit; hosp: hospital; IC: intensive care; ICU: intensive care unit; LOS: length of stay; no.: number; OCC: occupancy; pt: patient; SD: standard deviation; UK: United Kingdom; USA: United States of America.

***Thompson & Spiers, 1998*[14]**

1. Census- midnight
2. Calendar date- a bed occupancy day was calculated if a patient occupied a bed during any part of a calendar day
3. Day-to-day- a bed occupancy day was calculated if a patient occupied a bed from one day to the next or only part of a day
4. Hourly occupancy- exact number of hours

***Ridley & Rowan, 1997*[13]**

1. Census- midnight (mean & SD of daily proportions)
2. Census- 6am (mean & SD of daily proportions)
3. Census- midday (mean & SD of daily proportions)
4. Duration of admission for all patients admitted during the study period using ICU admission & discharge dates (annual proportion)
5. Duration of admission for all patients admitted during the study period using ICU admission & discharge d ates & times & rounded up to the nearest whole day (annual proportion)
6. Duration of admission for all patients admitted during the study period using ICU admission & discharge dates & times (annual proportion)
7. Duration of admission each day using ICU admission & discharge dates & times (mean & SD of daily proportions), for patients treated during the study period

*SD= Standard Deviation

Fig. 2. Occupancy calculation methods.

information such as facility characteristics, utilisation data, Medicare data and a financial statement.[15] Data reported in the HCRIS uses funded rather than operational beds.[11,15,16] This underestimates actual occupancy as beds can be taken out of service for numerous reasons including cleaning, maintenance, patient isolation, and temporary changes in staff numbers as a result of staff leave or deployment.[1,18,19]

This underestimation of occupancy levels can be compounded by the use of calculation methods that miss some patients. Patients admitted and discharged from the ICU on the same day may be missed if duration of stay is being calculated using admission and discharge dates only. Similarly, the census methods for calculating occupancy can omit patients with short durations of stay.[13] In particular, midnight census is likely to produce the lowest occupancy level of the day as patients who are admitted in the morning and discharged at night will not be included.[1] The omission of these patients is considered to be so great that a unit with a midnight census of 87% is likely to have peak occupancy during the day of approximately 95%.[1] This indicates the sensitivity of the census method to local admission and discharge practices and the time of day it is calculated.

Some occupancy calculation methods can also lead to an overestimation of occupancy. This can be seen in methods that round patient duration of stay in the ICU up to the nearest whole day such as calendar date, day-to-day methods and those that use admission and discharge dates only to calculate patient duration of stay.[13] The census method can also overestimate occupancy, depending on the local admission and discharge practices and the time chosen for calculating occupancy. Using the midday census for example, a

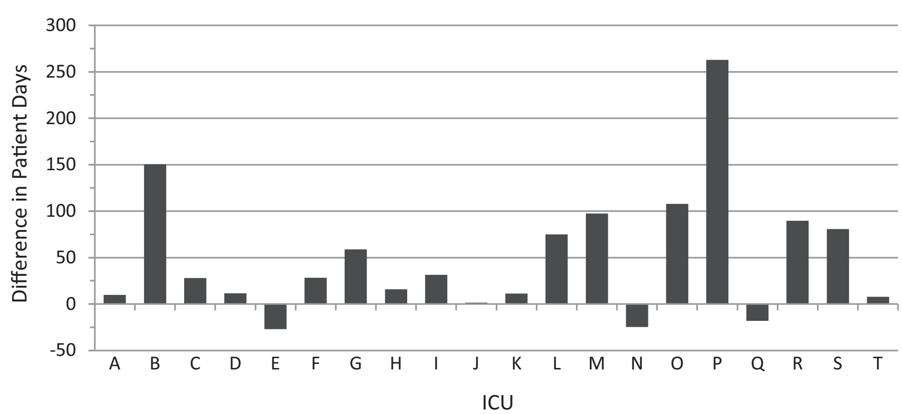

Fig. 3. Difference in patient days calculated by Ridley and Rowan[13] using methods six and seven. *Note*: As method seven was the reference method, a positive difference in patient days means method six overestimated patient days, whereas a negative difference means method six underestimated patient days.

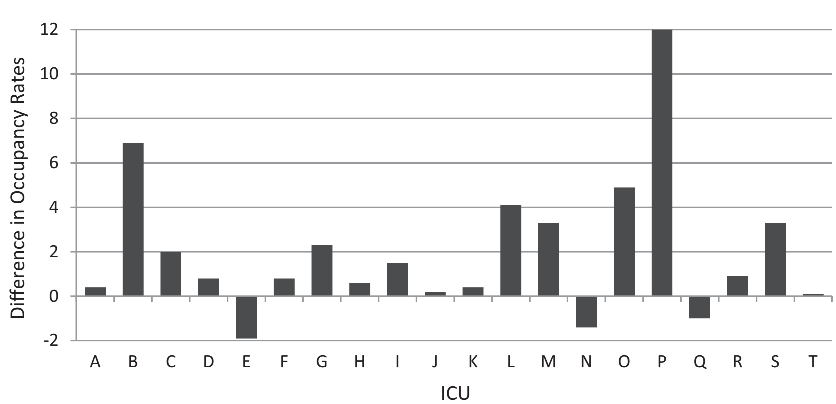

Fig. 4. Difference in occupancy rates calculated by Ridley and Rowan[13] using methods six and seven. *Note*: As method seven was the reference method, a positive difference in occupancy rates means method six overestimated occupancy, whereas a negative difference means method six underestimated occupancy.

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patient admitted to ICU at 11:30 am on day one and discharged at 12:30 pm on day two would be included in the occupancy calculation on both days despite their duration of stay being just over one 24-h period.[13]

Reporting

ICU occupancy levels are typically reported as yearly averages. The existence of local daily, monthly and seasonal patterns in ICU occupancy however, make average annual occupancy figures inadequate for policy and planning decisions and service evaluation.[1,8,10,20] The inadequacy of average annual occupancy can be seen in the significant variation between peak daytime occupancy and the midnight census.[1,13] Occupancy has also been shown to be considerably lower on weekends and over holiday periods as fewer elective procedures are typically scheduled at those times.[1,21] Additionally, there are peaks and troughs throughout the year with occupancy typically lower over the summer months.[1,9] Failure to account for variability in demand, particularly periods of higher than average occupancy, could significantly underestimate the capacity that units require to operate at a chosen target occupancy level.[10,20] Displaying occupancy rates using monthly data points for example, can indicate extended periods of higher occupancy (e.g. winter flu season) that may be masked by yearly average occupancy.[1] An understanding of the variations that can occur in occupancy over time allows for more accurate planning and unit management.[1]

Optimal occupancy

Findings from the literature suggested that high occupancy rates have a negative effect on patient outcomes and restrict access to ICU services. High occupancy rates, particularly those above 80%, were seen to have a direct relationship with ICU and hospital mortality and ICU readmission within seven days of discharge.[6,9] As occupancy increases there may also be escalating challenges of admitting patients into the ICU. Some of the consequences identified anecdotally in the literature include: refused or delayed admissions to ICU; cancellation of elective surgery; increased severity of illness on later admission to the ICU; nursing critically ill patients elsewhere in the hospital; transfer of emergency patients; hospital diversions; and premature discharge.[1,10,20]

There currently does not appear to be definitive evidence on what the optimal target for ICU occupancy should be however, it could be around 70–75%.[1,17,22] This is based on the premise that 100% occupancy is unobtainable as the discharge of existing patients and subsequent admission of new patients is not instantaneous and time is required for servicing and resetting the bed space.[13] Only three papers suggested an optimal occupancy target: one drawn from consensus finding of 47 intensive care physicians, the majority from Europe and the United Kingdom[22]; one using queuing analysis to estimate bed unavailability in ICUs[1]; and the third paper based its recommendation on existing national guidelines and one of the other papers included in this review.[17] It has been argued that a target of around 75–80% may be unrealistic with one study revealing that all the participating ICUs, even the largest units with the highest occupancy rates, operated at a lower occupancy than the recommended target.[16] Similarly, a study that examined peak daily ICU occupancy within three Canadian health regions over a seven year period also suggested an 80% target occupancy was too high.[23] Their findings indicated that providing enough ICU capacity to accommodate the peak daily occupancy 95% of the time would result in units operating at an average annual occupancy lower than 80% in all regions.[23]

Discussion

Key findings

The findings from this literature review demonstrated a lack of operational definitions for ICU occupancy and inconsistencies in calculation methods that are largely dependent on available data. Even when using the same data, the different calculation methods produced wide variation in occupancy rates due to the inclusion and exclusion criteria used. Some calculation methods underestimate occupancy levels (e.g. not factoring in unavailable beds), others can overestimate them (e.g. rounding patient duration of stay to the nearest whole day). When reviewing the literature for information on how to report, interpret and use occupancy figures and identifying optimal occupancy levels, it became apparent there were a number of related issues that required further discussion – these are outlined below.

Further considerations

Numerous challenges to setting optimal occupancy levels have been identified – particularly at the local level where an array of factors both intrinsic and extrinsic to the ICU, can impact on calculation methods and their interpretation. At the local ICU level, using overall occupancy rates mask the diversity of ICU patient casemix and are therefore inadequate for planning and managing ICU services.[6,7,10,20] It has been suggested for example, that high level occupancy should be differentiated from low level occupancy.[7,24] High level occupancy was defined as ICU care involving "interventions such as artificial ventilation, haemodialysis, vasoactive drugs or by risk of mortality combined with intensive care interventions or even combined with diagnosis".[24(p293)] Accordingly, low level care was care not involving these types of interventions or risk of mortality.[7,24] These concepts may be loosely equated to the Australian use of intensive care and high dependency beds respectively. Examining high and low level occupancy would allow units as well as service planners to develop an understanding of whether high-facility (i.e. ICU) beds are being used for patients requiring low-level care and whether low-facility (i.e. high dependency) beds are being used for patients requiring high-level care.[7] This would consequently allow for informed decisions to be made around resourcing ICUs adequately, and ensuring appropriateness of care.

Unit and hospital size are additional factors that require consideration when determining optimal occupancy targets. There is evidence to suggest that as hospital size increases, ICU occupancy also increases.[16] It was argued that smaller ICUs and those located in smaller or rural hospitals need to operate at lower occupancies because they do not have the economies of scale of larger units, i.e. they require more surge capacity proportionally and rural hospitals are less likely to be in close proximity to other facilities.[1,16] It has been recommended that ICUs with less than 5 beds should aim to operate at an occupancy rate of 45%.[1] There are no reported evaluations of this in Australian ICUs.

Factors extrinsic to the ICU that impact on occupancy levels include: the presence of a high dependency unit (HDU) or step-down facility; models of care; and bed management practices. Of the articles included in this review, two were specifically concerned with the impact of a HDU on the workload and occupancy of an ICU in the same hospital,[21–25] while others also touched on the issue.[8,15] Some research has suggested that the presence of HDUs or other step down units can reduce demand for intensive care services and marginally lower ICU occupancy.[15,21] In certain circumstances however, this reduction in occupancy may be disproportionate to the amount of capacity added to critical care services as a whole. One study revealed that when an additional 71% capacity was added to critical care services by opening a HDU, there was only a 16%

reduction in critical care occupancy.[8] It has therefore been suggested that falls in occupancy resulting from increased critical care capacity are transient and that with time, new areas of demand for intensive care services develop, resulting in increased occupancy levels.[8,21] Despite this, the use of the ICU becomes more appropriate with studies revealing a lower occupancy of the ICU by high dependency status patients and increased ICU occupancy by intensive care status patients.[8,25] Therefore, although the presence of a HDU may not serve to markedly lower the occupancy of the ICU or combined critical care services, it may improve the efficiency and appropriateness of service delivery which is an important consideration for health planners.[8,21]

Different models of care in the hospital have also been shown to impact on demand for ICU services, which may initially lower ICU occupancy. Some examples of these models include fast-track anaesthesia processes such as using propofol rather than high dose opioids which has led to shorter recovery times and a reduction in time to extubation.[26] Additionally, enhanced post-anaesthesia care unit (PACU) capabilities allows surgical patients who did not experience any complications to be cared for in the PACU and then discharged to the ward without needing to be admitted to the ICU.[27] A reduction in patients requiring admission to the ICU can also be seen with increased use of non-invasive ventilation in wards outside of the ICU.[15]

Some hospitals use bed management practices that can affect ICU occupancy figures. When HDU or appropriate ward beds are full for example, ICU patients who are deemed ready for a downgrade in care will continue to be nursed in the ICU.[1,10] Notably, these exit-blocked patients are included in occupancy calculations, highlighting the need to consider a number of other hospital level measures in conjunction with occupancy, such as exit and access (i.e. the number of patients unable to be admitted to ICU) block, no-bed transfers, length of wait for admission into ICU, and the admission and discharge practices of each unit and hospital.

It has also been recognised that occupancy is only one measure of unit activity and that it should be considered in conjunction with other unit level information. As two units could have the same occupancy figure but very different levels of patient throughput and length of stay (LOS), these should be examined in conjunction with occupancy in order to develop a comprehensive picture of unit activity.[13] Methods used to calculate measures of unit activity however, must also be standardised if comparisons are to be made.[13]

Implications for practice

There are a number of implications for ICU occupancy research, policy and planning arising from this literature review. These include further operationalisation of the definition of occupancy, detail pertaining to how the denominator should be calculated, patient inclusion and exclusion criteria, occupancy rates calculated by patient type, and the time period over which occupancy should be examined and reported.

Variation in calculation methods typically result because of differences in available data internationally and for individual ICUs. After reviewing the literature pertaining to occupancy calculation methods, the following method is recommended for use across the board.

$$\text{Occupancy} = \frac{\text{Total number of patient bed hours}}{\text{Total number of available bed hours}}$$

In this definition, 'bed hours' can be as precise as the data collection system allows i.e. for ICUs that have clinical information systems, the time can be calculated to the second. Importantly, accurate calculation of occupancy must be focused on beds rather than patient admissions. If these data are not available, occupancy can be calculated from patient admissions data by using the sum of the duration of admissions for each patient *admitted* during the time period for which occupancy is being calculated. There are however, limitations to this approach that compromise the accuracy of occupancy figures: (1) patients admitted prior to the study period but who occupied a bed in the unit during that time are excluded; and (2) parts of patient stays that extend beyond the study period are included as long as the patient was admitted during the study period.[13] In contrast, a focus on beds and the number of hours a bed is occupied reflects the actual patient occupancy of an ICU for any given time period.

For units with accurate and precise real-time data, the following beds should be excluded from the denominator of the occupancy calculation:

- physical beds that the unit is not funded to staff;
- temporarily unavailable beds due to cleaning after patient discharge;
- closed beds due to maintenance; and
- closed beds due to staff leave or deployment.

When evaluating efficiency of resource use, it would also be beneficial to quantify the above exclusions.

When using reported occupancy figures and comparing international studies it is important to consider which, if any, beds have been excluded from the calculation of the denominator. For example, occupancy figures from the United States using HCRIS data use funded beds so do not exclude any of the above categories.[11,15,16] In contrast, CCRS (Critical Care Resource System) data used in NSW, Australia excludes unavailable beds (Zhang, A [NSW Intensive Care Coordination and Monitoring Unit] Conversation with: Laura Tierney [NSW Intensive Care Coordination and Monitoring Unit] 2012 May 11). It is likely that there is currently great diversity in the way the denominator is calculated irrespective of geographical location, therefore caution must be used when making such comparisons.

In addition to overall occupancy, ICU occupancy by different patient types should also be examined to give a more comprehensive understanding of unit activity. If accurate data is collected on the number of bed hours for ICU status patients, HDU status patients and exit blocked patients, the proportion of unit occupancy contributed by these groups should also be considered.

The typical variations in occupancy mean that it is appropriate to calculate daily occupancy, particularly at the local ICU level. This allows for an observation of variations within and between weeks, months and years. For planning and evaluation activities however, it may be more appropriate to calculate and display monthly variation in occupancy rates to allow for comparisons and the detection of seasonal trends.[8,9,21]

The lack of good quality evidence for optimal ICU occupancy makes it unclear what target should be set. It does appear however, that there are critical points at which occupancy levels become problematic. In addition to the findings reported previously suggesting that high occupancy rates impact negatively on patient outcomes,[6,9] a study that used queuing theory to model patient flow and resource use in a paediatric ICU found that the highest safe occupancy level for the unit was 80–85%, above which hospital diversions and transfers increased abruptly.[18] In another study focused on delayed discharge from an adult ICU, an increased number of delays were observed when occupancy was between 70% and 80%.[28] In the absence of definitive evidence, a pragmatic approach would be to develop target or optimal occupancy levels that are informed and driven by acceptable standards in clinical outcomes and service provision.[1]

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It is clear there is a need to develop more specific guidelines that account for the unique characteristics of each ICU, and strategies such as lower occupancy targets for smaller units[1] have been proposed. However, NSW CCRS data indicates that many small and rural ICUs typically operate at occupancies higher than the suggested 45% target (Zhang, A [NSW Intensive Care Coordination and Monitoring Unit] Conversation with: Laura Tierney [NSW Intensive Care Coordination and Monitoring Unit] 2012 May 11). This may be due to different bed management practices in smaller hospitals where the ICU occupancy may consist of ICU and HDU status patients, as well as stable patients with complex nursing requirements. Occupancy may therefore need to be evaluated differently for smaller ICUs, such as calculating unit occupancy for intensive care status patients only. This issue requires further research and evaluation.

Limitations

Despite the insights gained from this literature review there are some limitations. Firstly, it was not conducted as a systematic review however the literature searching methods provided extensive literature coverage. Secondly, due to the lack of comprehensive and internationally focused ICU occupancy research, the applicability of findings across different ICUs may be limited. There are many and varied models of delivering intensive and critical care services internationally, with differences in definition of intensive care beds,[29] staffing profiles[24] admission policies,[24,30] ICU organisation within the hospital e.g. "closed" versus "open" units,[24] and gradation of ICU care e.g. presence and function of HDUs.[24,30] These differences make it difficult to translate findings from international studies to different clinical contexts.

The ICU occupancy literature also had some collective limitations. Retrospective study designs were utilised in six of the 13 research studies reported in this review and were therefore restricted to calculating ICU occupancy using the available data. The calculation method used for measuring occupancy varied greatly between studies, making comparisons difficult. There was also a lack of good quality evidence pertaining to reporting methods and optimal ICU occupancy targets.

Further research

The results of this literature review have highlighted gaps in the literature and the need for more research in the area. The lack of relevant literature in Australia highlights the need for further research that takes the Australian context of ICU and critical care into consideration. This specific context may result in the determination of a different set of occupancy targets for ICUs. Another unknown factor is the frequency and length of time beds are unavailable (e.g. re-setting the bed space, staff shortages) and what impact this has on occupancy measurement.

The need for further research into the intrinsic and extrinsic factors impacting optimal occupancy is also apparent. Although some strategies have already been suggested for taking these factors into consideration, further work is required to determine more precisely the impact of these on unit occupancy and how they should be used to: (1) calculate more specific occupancy targets for each ICU; and (2) ensure optimal occupancy levels are maintained. The immediate and long term effects of step-down facilities in the hospital and alternate models of care on ICU occupancy also need to be explored. Finally, there is a lack of evidence on the impact of high occupancy in adult ICUs on measures of service delivery both in the ICU and more broadly across the hospital. An understanding of this relationship is important as it would reinforce the need to maintain ICU occupancy below defined optimal target levels.

Conclusion

This literature review has explored occupancy operationalisation and measurement, issues surrounding occupancy reporting, use, and interpretation, and the determination of optimal occupancy levels for ICUs. There was significant variation in the calculation of occupancy and although it has not been widely tested, the proposed measurement method has been drawn from existing work. It is evident that average annual ICU occupancy must not be considered in isolation. The array of factors discussed that impact ICU occupancy include: daily, weekly and monthly variations in occupancy; unit level factors such as patient case-mix, LOS, throughput and unit size; and factors outside of the ICU such as hospital size, presence of step-down facilities in the hospital, models of care and bed management practices. Despite there being no definitive evidence pertaining to optimal occupancy levels, findings from the collective literature suggest poorer patient outcomes and negative impacts on service delivery from around the 70–75% mark, and this should be factored into future service evaluations. Although it is recognised in the literature that ICU occupancy is impacted by many intrinsic and extrinsic factors, the specific effect they have on the maintenance of optimal ICU occupancy cannot currently be determined. More research is required in all areas concerning ICU occupancy, particularly in Australia.

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Please cite this article in press as: Tierney LT, Conroy KM. Optimal occupancy in the ICU: A literature review. *Aust Crit Care* (2014), http://dx.doi.org/10.1016/j.aucc.2013.11.003

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